

Scope of the Solution

The proposed Dam Water Level Alert System is designed as a prototype to demonstrate automated water level monitoring using an Arduino Uno. An HC-SR04 ultrasonic sensor continuously measures the distance between the sensor and the water surface. Based on the measured distance, the system classifies the water level into three zones: Safe, Warning, and Danger.

The current water level is displayed on a 16×2 LCD, while Green, Yellow, and Red LEDs provide visual indication of the water level status. A buzzer is activated when the water reaches the danger level, and a servo motor simulates the automatic opening of a flood gate.

This project demonstrates the basic principles of real-time monitoring and alert generation using embedded systems. It is intended for educational purposes and can be extended in the future by integrating IoT-based remote monitoring, cloud data logging, GSM alerts, or automatic dam gate control.